An official website of the United States government
Here's how you know

NIH Websites Are Changing

- Get the Latest Updates and FAQs on the transition
- Per Executive Order 14434, federal agencies are now using "Super Intelligence" (SI) in place of "Artificial Intelligence" (AI). You may see both terms across NIH content during this transition.

NM_000298.6(PKLR):c.1456C>T (p.Arg486T...

Germline

Classification

Pathogenic/Likely pathogenic
Pathogenic (26); Likely pathogenic (4)
30 out of 33 submissions contributed to this classification

Somatic

No data submitted for somatic clinical impact

Somatic

No data submitted for oncogenicity

Variant Details

Variant details

Identifiers: NM_000298.6(PKLR):c.1456C>T (p.Arg486Trp)

Variation ID: 1513 Accession: VCV000001513.102

Type and length: single nucleotide variant, 1 bp **Location:**

Cytogenetic: 1q22 1: 155291918 (GRCh38) [NCBI UCSC] 1: 155261709 (GRCh37) [NCBI UCSC]

Timeline in ClinVar:

	First in ClinVar	Last submission	Last evaluated
Germline	Oct 14, 2016	Oct 3, 2026	Mar 20, 2026

HGVS:

Nucleotide	Protein	Molecular consequence
NM_000298.6:c.1456C>T MANE SELECT	NP_000289.1:p.Arg486Trp	missense
NM_181871.4:c.1363C>T	NP_870986.1:p.Arg455Trp	missense
NC_000001.11:g.155291918G>A		

... more HGVS

Protein change: R486W, R455W **Other names:** p.(Arg486Trp)

Canonical SPDI: NC_000001.11:155291917:G:A

Global minor allele frequency (GMAF) : 0.00160 (A) **Allele frequency :**

1000 Genomes Project 0.00160

1000 Genomes Project 30x 0.00172

NHLBI Exome Sequencing Project (ESP) Exome Variant Server 0.00208

Trans-Omics for Precision Medicine (TOPMed) 0.00229

The Genome Aggregation Database (gnomAD) 0.00270

The Genome Aggregation Database (gnomAD) 0.00279

Exome Aggregation Consortium (ExAC) 0.00285

The Genome Aggregation Database (gnomAD), exomes 0.00297

The Genome Aggregation Database (gnomAD), exomes 0.00305

Links :

ClinGen: CA215080
UniProtKB: P30613#VAR_004066
OMIM: 609712.0009
dbSNP: rs116100695
VarSome

Feedback

Genes

Gene	OMIM	ClinGen Gene Dosage Sensitivity Curation		Variation Viewer	Related variants	
		HI score	TS score		Within gene	All
PKLR		-	-	GRCh38 GRCh38 GRCh37	482	513

Conditions - Germline

Condition	Classification (# of submissions)	Review status	Last evaluated	Variation/condition record
Pyruvate kinase deficiency of red cells	Pathogenic/Likely pathogenic (15)	★★☆☆	Dec 11, 2025	RCV000001577.50
not provided	Pathogenic/Likely pathogenic (11)	★★☆☆	Feb 19, 2026	RCV000724015.82
Pyruvate kinase deficiency of red cells Pyruvate kinase hyperactivity	Pathogenic (1)	★☆☆☆	May 28, 2022	RCV000762857.11
Pyruvate kinase hyperactivity	Pathogenic (2)	★☆☆☆	May 4, 2022	RCV000991156.10
See cases	Pathogenic (1)	★☆☆☆	Nov 13, 2020	RCV002251854.9
PKLR-related disorder	Pathogenic (2)	★★☆☆	Mar 20, 2026	RCV004751189.3
Inborn genetic diseases	Pathogenic (1)	★☆☆☆	Feb 23, 2026	RCV006659115.1

Submissions - Germline

Classification (Last evaluated)	Review status (Assertion criteria)	Condition	Submitter	Collapse all rows
Pathogenic (May 04, 2022) C Contributing to aggregate classification	★☆☆☆ (Mendelics Assertion Criteria 2019)	Pyruvate kinase hyperactivity	Mendelics Accession: SCV002518869.1 First in ClinVar: May 28, 2022 Last updated: May 28, 2022	Observation: 1 • Collection method: clinical testing • Allele origin: germline • Affected status: unknown
Pathogenic (Mar 04, 2025) C Contributing to aggregate classification	★☆☆☆ (ACMG Guidelines, 2015)	not provided	Center for Genomic Medicine, Rigshospitalet, Copenhagen University Hospital Accession: SCV004243293.3 First in ClinVar: Feb 14, 2024 Last updated: Mar 11, 2025	

Publications:
PubMed: 11960989, 17360088

Observation 1

Collection method: clinical testing
Allele origin: germline
Affected status: unknown

Likely pathogenic (Apr 14, 2017) C Contributing to aggregate classification	★☆☆☆ (EGL Classification Definitions 2015)	not provided	Eurofins Ntd Llc (ga) Accession: SCV000331623.5 First in ClinVar: Dec 06, 2016 Last updated: Apr 13, 2025	

Classification (Last evaluated)	Review status (Assertion criteria)	Condition	Submitter	Collapse all rows

Other databases

http://www.egl-eurofins.com/emvclass/emvclass.php?approved_symbol=PKLR

Observation 1

Collection method: clinical testing
Allele origin: germline
Affected status: unknown
Number of individuals with the variant: 8
Zygosity: 8 Single Heterozygotes
Sex: mixed

Pathogenic
(May 28, 2022)

C Contributing to aggregate classification

★★★★
(ACMG Guidelines, 2015)

Pyruvate kinase hyperactivity
Pyruvate kinase deficiency of red cells

Fulgent Genetics, Fulgent Genetics
Accession: SCV000893217.2
First in ClinVar: Mar 31, 2019
Last updated: Dec 31, 2022

Publications:

PubMed: 25741868

DOI: 10.1038/gim.2015.30
pmc: 4544753

Observation 1

Collection method: clinical testing
Allele origin: unknown
Affected status: unknown

Likely Pathogenic
(Dec 23, 2021)

C Contributing to aggregate classification

★★★★
(ACMG Guidelines, 2015)

Pyruvate kinase deficiency of red cells
(Autosomal recessive inheritance)

Laboratory for Molecular Medicine, Mass General Brigham Personalized Medicine
Accession: SCV000713497.4
First in ClinVar: Apr 09, 2018
Last updated: Apr 20, 2024

Publications:

PubMed: 11054094, 11328279, 11960989, 18759866, 26728349, 27346685, 27871768, 28133914, 29396846, 29555771, 30609409, 31980526, 32036089, 32974842, 33631127, 8483951, 9827908

Comment:

The p.Arg486Trp variant in PKLR has been reported in >30 compound heterozygous individuals with pyruvate kinase deficiency and segregated in 1 affected relative (PKD; Baronciani 1993 PMID:8483951, Zarza 1998 PMID:9827908, Manco 2000 PMID:11054094, Zanella 2001 PMID:11328279, Kedar 2009 PMID:18759866, Kager 2016 PMID:26728349, Jaouani 2017 PMID:28133914, Canu 2020 PMID:32974842, Milanesio 2021, Jamwal 2020 PMIDL32036089). It has also been identified in 0.812% (204/25114) of Finnish (unknown frequency of PKD) and 0.29% (375/128890) of European chromosomes (1:20,000 frequency of PKD) by gnomAD (http://gnomad.broadinstitute.org). This variant has been reported in ClinVar (Variation ID 1513). Enzymatic studies suggest that the p.Arg486Trp variant may be a mild mutation with reduced activity (Zarza 1998 PMID:9827908, Zanella 2001 PMID:11328279, Valentini 2002 PMID:11960989, and Kedar 2009 PMID:18759866). In summary, although additional studies are required to fully establish its clinical significance, the p.Arg486Trp variant meets criteria to be classified as likely pathogenic for autosomal recessive pyruvate kinase deficiency. ACMG/AMP Criteria applied: PM3_VeryStrong, PS3_Moderate. (less)

Observation 1

Collection method: clinical testing
Allele origin: germline
Affected status: unknown

Pathogenic
(Dec 30, 2024)

C Contributing to aggregate classification

★★★★
(ACMG Guidelines, 2015)

not provided

Revvity Omics, Revvity
Accession: SCV002024662.4
First in ClinVar: Nov 29, 2021
Last updated: Sep 06, 2025

Observation: 1

- Collection method: clinical testing
- Allele origin: germline
- Affected status: unknown

Pathogenic
(Apr 14, 2025)

C Contributing to aggregate classification

★★★★
(ACMG Guidelines, 2015)

Pyruvate kinase deficiency of red cells
(Autosomal recessive inheritance)

First Genomix Gene Laboratory, Genetic Diagnostics Department
Accession: SCV007095982.1
First in ClinVar: Nov 08, 2025
Last updated: Nov 08, 2025

Classification (Last evaluated)	Review status (Assertion criteria)	Condition	Submitter	Collapse all rows

Comment:

As part of Carrier Screening testing performed at First Genomix, this variant was identified in a heterozygous state in a patient who is not affected with this condition. (less)

Observation 1

Collection method: clinical testing
Allele origin: germline
Affected status: no
Number of individuals with the variant: 1
Secondary finding: yes

Pathogenic
(-)

C Contributing to aggregate classification

★☆☆☆
(ACMG Guidelines, 2015)

Pyruvate kinase deficiency of red cells

Juno Genomics, Hangzhou
Juno Genomics, Inc
Accession: SCV005415943.3
First in ClinVar: Nov 30, 2024
Last updated: Mar 01, 2026

Comment:

For recessive disorders, detected in trans with a pathogenic variant.;Co-segregation with disease in multiple affected family members in a gene definitively known to cause the disease.;Well-established in vitro or in vivo functional studies supportive of a damaging effect on the gene or gene product. (less)

Observation 1

Collection method: clinical testing
Allele origin: germline
Affected status: yes

Likely pathogenic
(Jan 27, 2016)

C Contributing to aggregate classification

★☆☆☆
(ACMG Guidelines, 2015)

Pyruvate kinase deficiency of red cells

Knight Diagnostic Laboratories, Oregon Health and Sciences University
Study: CSER-NextGen
Accession: SCV000223943.2
First in ClinVar: Oct 05, 2015
Last updated: Apr 20, 2017

Comment:

The c.1456C>T (p.Arg486Trp) (NM_ 000298.5) missense variant has been reported in several unrelated individuals diagnosed with PK deficiency. These families have a well-documented clinical history of anemia (Zarza et al. 1998). Case-control studies have reported this variant as a common disease causing variant accounting for 9%-32% of diagnosed cases (Zarza et al., 1998; Zanella et al., 2001; Kedar et al., 2009). This variant has often been reported in trans with several pathogenic variants, and functional studies have shown that patients harboring this variant, in a compound heterozygous state, have reduced PK activity relative to normal controls (Zarza et al., 1998; Zanella et al., 2001; Valentini et al. 2002; Kedar et al., 2009). This c.1456C>T variant is reported at low frequency in the population databases (ESP = 0.291%; 1000 Genomes = 1%; ExAC = 0.298%), and multiple in silico algorithms predict this variant to have a deleterious effect (GERP = 4.85; CADD = 18.75; PolyPhen = 1; SIFT = 0). Therefore, this collective evidence supports the classification of the c.1456C>T (p.Arg486Trp) variant as a recessive Likely Pathogenic variant for PK deficiency. (less)

Observation 1

Collection method: clinical testing
Allele origin: germline
Affected status: no

Pathogenic
(Nov 13, 2020)

C Contributing to aggregate classification

★☆☆☆
(ACMG Guidelines, 2015)

See cases

Laboratorio de Genetica e Diagnostico Molecular, Hospital Israelita Albert Einstein
Accession: SCV002523854.1
First in ClinVar: Jun 11, 2022
Last updated: Jun 11, 2022

Classification (Last evaluated)	Review status (Assertion criteria)	Condition	Submitter	Collapse all rows

Comment:
ACMG classification criteria: PS3, PS4, PM3, PP1, PP3, PP4, BS2 (less)

Observation 1

Collection method: clinical testing
Allele origin: germline
Affected status: yes
Clinical Features:
Autosomal recessive inheritance (present)
Geographic origin: Brazil

Pathogenic
(Nov 10, 2023)
C Contributing to aggregate classification

★★★★
(LabCorp Variant Classification Summary - May 2015)

Pyruvate kinase deficiency of red cells

Women's Health and Genetics/Laboratory Corporation of America, LabCorp
Accession: SCV004223638.1
First in ClinVar: Jan 06, 2024
Last updated: Jan 06, 2024

Publications:
PubMed: 11960989, 16704447, 9160692, 9482576

Comment:
Variant summary: PKLR c.1456C>T (p.Arg486Trp) results in a non-conservative amino acid change located in the Pyruvate Kinase C terminal domain (IPR015795) of the encoded protein sequence. Five of five in-silico tools predict a damaging effect of the variant on protein function. The variant allele was found at a frequency of 0.003 in 251132 control chromosomes in the gnomAD database, including 2 homozygotes. c.1456C>T has been reported in the literature in multiple homozygous and compound heterozygous individuals affected with moderate to severe Pyruvate Kinase Deficiency Of Red Cells (examples: Zanella_1997, Pastore_1998, Pissard_2006). These data indicate that the variant is very likely to be associated with disease. One publication reports experimental evidence that this variant moderately affects PKLR function (example: Valentini_2002). Fifteen submitters have cited clinical-significance assessments for this variant to ClinVar after 2014. All submitters classified the variant as pathogenic/likely pathogenic. Based on the evidence outlined above, the variant was classified as pathogenic. (less)

Observation 1

Collection method: clinical testing
Allele origin: germline
Affected status: unknown

Pathogenic
(Mar 29, 2024)
C Contributing to aggregate classification

★★★★
(ACMG Guidelines, 2015)

Pyruvate kinase deficiency of red cells

Genomic Medicine Center of Excellence, King Faisal Specialist Hospital and Research Centre
Accession: SCV004808207.1
First in ClinVar: Apr 06, 2024
Last updated: Apr 06, 2024

Observation: 1
- Collection method: clinical testing
- Allele origin: germline
- Affected status: unknown

Pathogenic
(Jul 05, 2022)
C Contributing to aggregate classification

★★★★
(ACMG Guidelines, 2015)

not provided

Clinical Genetics Laboratory, Skane University Hospital Lund
Accession: SCV005197175.1
First in ClinVar: Aug 25, 2024
Last updated: Aug 25, 2024

Observation: 1
- Collection method: clinical testing
- Allele origin: germline
- Affected status: yes

Pathogenic
(Mar 31, 2022)
C Contributing to aggregate classification

★★★★
(ACMG Guidelines, 2015)

Pyruvate kinase deficiency of red cells
(Autosomal recessive inheritance)

Victorian Clinical Genetics Services, Murdoch Childrens Research Institute
Accession: SCV005399102.1
First in ClinVar: Nov 24, 2024
Last updated: Nov 24, 2024

Publications:
PubMed: 16704447, 32036089, 34093240

Comment:
Based on the classification scheme VCGS_Germline_v1.3.4, this variant is classified as pathogenic. The following criteria are met: 0102 - Loss of function is a known mechanism of disease in this gene and is associated with pyruvate kinase deficiency (MIM#266200). (I) 0108 - This gene is associated with both recessive and dominant disease. Both dominant elevated adenosine triphosphate of erythrocytes (MIM#102900) and recessive pyruvate kinase deficiency (MIM#266200) have been associated with PKLR. However, most of the literature reports associate with the recessive condition. (I) 0200 - Variant is predicted to result in a missense amino acid change from arginine to tryptophan. (I) 0251 - This variant is heterozygous. (I) 0304 - Variant is present in gnomAD (v2) <0.01 for a recessive condition (829 heterozygotes, 2 homozygotes). (SP) 0309 - An alternative amino acid change at the same position has been observed in gnomAD (v2) (10 heterozygotes, 0 homozygotes). (I) 0501 - Missense variant consistently predicted to be damaging by multiple in silico tools and poorly conserved with a major amino acid change. (SP) 0600 - Variant is located in the annotated pyruvate kinase C-terminal domain (DECIPHER). (I) 0801 - This variant has strong previous evidence of pathogenicity in unrelated individuals. This variant is one of the most common variants reported in individuals with pyruvate kinase deficiency (ClinVar, PMIDs: 16704447, 32036089, 34093240). (SP) Legend: (SP) - Supporting pathogenic, (I) - Information, (SB) - Supporting benign (less)

Observation 1

Collection method: clinical testing

Classification (Last evaluated)	Review status (Assertion criteria)	Condition	Submitter	Collapse all rows

Allele origin: germline
Affected status: unknown

Pathogenic
(May 20, 2023)

C Contributing to aggregate classification

★☆☆☆
(ACMG Guidelines, 2015)

Pyruvate kinase deficiency of red cells
(Autosomal recessive inheritance)

Neuberg Centre For Genomic Medicine, NCGM
Accession: SCV004171978.5
First in ClinVar: Dec 09, 2023
Last updated: Jan 25, 2025

Comment:

The observed missense c.1456C>T(p.Arg486Trp) variant in PKLR gene has been reported previously in homozygous or compound heterozygous state in individual(s) affected with Pyruvate kinase deficiency (PKD) (Dongerdiye et al., 2023). Experimental studies have shown that this variant moderately affects PKLR function (Valentini et al., 2002). This variant is reported with the allele frequency of 0.3% in the gnomAD Exomes. This variant has been reported to the ClinVar database as Likely Pathogenic / Pathogenic (multiple submitters). The amino acid Arg at position 486 is changed to a Trp changing protein sequence and it might alter its composition and physico-chemical properties. The amino acid change p.Arg486Trp in PKLR is predicted as conserved by GERP++ and PhyloP across 100 vertebrates. Multiple lines of computational evidence (Polyphen - Possibly Damaging, SIFT - Damaging, and MutationTaster - Disease causing automatic) predict a damaging effect on protein structure and function for this variant. For these reasons, this variant has been classified as Pathogenic. (less)

Observation 1

Collection method: clinical testing
Allele origin: germline
Affected status: yes
Clinical Features:
Abnormality of blood and blood-forming tissues (present)
Comment on clinical features:
Consanguinity: Present Investigations: HbH 16.8% increased Possible clinical diagnosis: Monocytic hypochromic hemolytic anemia, alpha thalassemia

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Pathogenic
(Jan 17, 2020)

C Contributing to aggregate classification

★☆☆☆
(Submitter's publication)

Pyruvate kinase deficiency of red cells

Genetics and Genomic Medicine Centre, NeuroGen Healthcare, NeuroGen Healthcare
Accession: SCV005873637.2
First in ClinVar: Mar 11, 2025
Last updated: Apr 13, 2025

Observation 1

Collection method: clinical testing
Allele origin: unknown
Affected status: no
Clinical Features:
Delayed speech and language development (present) , Global developmental delay (present) , Atypical behavior (present)
Comment on clinical features:
Neurodevelopmental disorder
Age: 0-9 years
Sex: male
Ethnicity/Population group: South East Asian
Geographic origin: Bangladesh
Secondary finding: yes

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pathogenic
(Jul 07, 2025)

C Contributing to aggregate classification

★☆☆☆
(ACMG Guidelines, 2015)

Pyruvate kinase deficiency of red cells
(Autosomal recessive inheritance)

Institute of Human Genetics, University of Leipzig Medical Center
Accession: SCV006324914.1
First in ClinVar: Sep 22, 2025
Last updated: Sep 22, 2025

Comment:

Criteria applied: PS4,PM3,PS3_SUP,PM1_SUP,PM5_SUP,PP3 (less)

Observation 1

Collection method: clinical testing
Allele origin: unknown
Affected status: yes
Clinical Features:
Fatigue (present) , Hepatosplenomegaly (present) , Elliptocytosis (present) , Spontaneous hemolytic crises (present) , Reduced red cell pyruvate kinase level (present)
Sex: male

Classification (Last evaluated)	Review status (Assertion criteria)	Condition	Submitter	Collapse all rows

Likely pathogenic
(Sep 10, 2025)
C Contributing to aggregate classification

★☆☆☆
(GeneDx Variant Classification Process June 2021)

not provided

GeneDx
Accession: SCV001792018.5
First in ClinVar: Aug 18, 2021
Last updated: Sep 22, 2025

Comment:

In silico analysis supports that this missense variant has a deleterious effect on protein structure/function; This variant is associated with the following publications: (PMID: 30681718, 29555771, 32279356, 37895316, 31980526, 29549173, 32165483, 11960989, 20981092, 8483951, 11054094, 24533562, 29349879, 29519373, 29519369, 9827908, 18759866, 27346685, 29396846, 30358897, 28133914, 32543069, 32043619, 30609409, 31974203, 32974842, 26728349, 33054133, 33054047, 23770304, 33054048, 34426522, 34662886, 11328279, 18708292, 32036089, 27871768, 9482576, 33631127, 37188156, 36892591, 38581917, 36825813, 35989577, 38434380, 37671043, 38811201) (less)

Observation 1

Collection method: clinical testing
Allele origin: germline
Affected status: yes

Pathogenic
(Dec 11, 2025)
C Contributing to aggregate classification

★☆☆☆
(ACMG Guidelines, 2015)

Pyruvate kinase deficiency of red cells
(Autosomal recessive inheritance)

Department of Molecular Genetics, Istishari Arab Hospital
Accession: SCV007122147.1
First in ClinVar: Dec 14, 2025
Last updated: Dec 14, 2025

Publications:

PubMed: 11054094, 11960989, 8483951

Comment:

The PKLR variant c.1456C>T p.(Arg486Trp) causes an amino acid change from Arg to Trp at position 486. According to HGMD Professional 2022.1, this variant has previously been described as disease-causing for Pyruvate kinase deficiency by Baronciani et al., 1993 (PMID: 8483951), Manco et al., 2000 (PMID: 11054094), Valentini et al., 2002 (PMID: 11960989). Additionally, this variant was previously reported in our lab in a case with abnormality of the upper limb, brain atrophy, generalized hypotonia, microcephaly, persistent head lag, and polycystic kidney dysplasia. It is classified as pathogenic (class 1) according to the recommendations of ACMG. (less)

Observation 1

Collection method: clinical testing
Allele origin: germline
Affected status: yes

Pathogenic
(Sep 05, 2025)
C Contributing to aggregate classification

★☆☆☆
(ACMG Guidelines, 2015)

not provided

Mayo Clinic Laboratories, Mayo Clinic
Accession: SCV001713042.5
First in ClinVar: Jun 15, 2021
Last updated: Jan 11, 2026

Comment:

PP3_moderate, PP4, PM1, PM3, PS4 (less)

Observation 1

Collection method: clinical testing
Allele origin: germline
Affected status: unknown
Number of individuals with the variant: 22

Pathogenic
(Mar 06, 2025)
C Contributing to aggregate classification

★☆☆☆
(ARUP Molecular Germline Variant Investigation Process 2024)

not provided

ARUP Laboratories, Molecular Genetics and Genomics, ARUP Laboratories
Accession: SCV000885966.9
First in ClinVar: Dec 15, 2018
Last updated: Jan 24, 2026

Comment:

The PKLR c.1456C>T; p.Arg486Trp variant (rs116100695, ClinVar Variation ID: 1513) is commonly reported in the literature as compound heterozygous variant in individuals with pyruvate kinase deficiency (Baronciani 1993, Baronciani 1995, Canu 2020, Jamwal 2020, Kedar 2009, Lenzner 1997, Manco 2000, Zanella 2001, Zarza 1998). This variant is found in the general population with an overall allele frequency of 0.3% (833/282,498 alleles, including two homozygotes) in the Genome Aggregation Database (v2.1.1.). Structural and functional analysis of the variant protein show a moderate decrease in the catalytic efficiency (Valentini 2002). Computational analyses predict that this variant is deleterious (REVEL 0.914). Based on available information, this variant is considered to be pathogenic. References: Baronciani L et al. Analysis of pyruvate kinase-deficiency mutations that produce nonspherocytic hemolytic anemia. Proc Natl Acad Sci U S A. 1993 May 1. PMID: 8483951. Baronciani L et al. Molecular study of pyruvate kinase deficient patients with hereditary nonspherocytic hemolytic anemia. J Clin Invest. 1995 Apr. PMID: 7706479. Canu G et al. Identification and in silico characterization of a novel PKLR genotype in a Turkish newborn. Mol Biol Rep. 2020 Oct. PMID: 32974842. Jamwal M et al. Next-Generation Sequencing-Based Diagnosis of Unexplained Inherited Hemolytic Anemias Reveals Wide Genetic and Phenotypic Heterogeneity. J Mol Diagn. 2020 Apr. PMID: 32036089. Kedar P et al. Spectrum of novel mutations

Classification (Last evaluated)	Review status (Assertion criteria)	Condition	Submitter	Collapse all rows

in the human PKLR gene in pyruvate kinase-deficient Indian patients with heterogeneous clinical phenotypes. Clin Genet. 2009 Feb. PMID: 18759866. Lenzner C et al. Molecular analysis of 29 pyruvate kinase-deficient patients from central Europe with hereditary hemolytic anemia. Blood. 1997 Mar 1. PMID: 9057665. Milanesio B et al. Six novel variants in the PKLR gene associated with pyruvate kinase deficiency in Argentinian patients. Clin Biochem. 2021 May. PMID: 33631127. Valentini G et al. Structure and function of human erythrocyte pyruvate kinase. Molecular basis of nonspherocytic hemolytic anemia. J Biol Chem. 2002 Jun 28. PMID: 11960989. Zanella A et al. Molecular characterization of the PK-LR gene in sixteen pyruvate kinase-deficient patients. Br J Haematol. 2001 Apr. PMID: 11328279. Zarza R et al. Molecular characterization of the PK-LR gene in pyruvate kinase deficient Spanish patients. Red Cell Pathology Group of the Spanish Society of Haematology (AEHH). Br J Haematol. 1998 Nov. PMID: 9827908. (less)

Observation 1

Collection method: clinical testing
Allele origin: germline
Affected status: unknown

Pathogenic
(Jan 24, 2026)

C Contributing to aggregate classification

★☆☆☆
(Invitae Variant Classification Sherloc (09022015))

not provided

Labcorp Genetics (formerly Invitae), Labcorp
Accession: SCV001035168.7
First in ClinVar: Dec 17, 2019
Last updated: Mar 07, 2026

Publications:

PubMed: 10354117, 11054094, 11328279, 11960989, 17360088, 17574881, 27871768, 32974842, 9057665, 9482576, 9657767, 9827908

Comment:

This sequence change replaces arginine, which is basic and polar, with tryptophan, which is neutral and slightly polar, at codon 486 of the PKLR protein (p.Arg486Trp). This variant is present in population databases (rs116100695, gnomAD 0.8%), and has an allele count higher than expected for a pathogenic variant. This missense change has been observed in individual(s) with pyruvate kinase deficiency (PMID: 9057665, 9657767, 9827908, 11054094, 17574881, 27871768, 32974842). In at least one individual the data is consistent with the variant being in trans (on the opposite chromosome) from a pathogenic variant. This variant is commonly associated with mild to asymptomatic disease when observed in the homozygous state. However, this variant in conjunction with another pathogenic variant is a common cause of mild chronic hemolytic anemia, particularly in populations of southern Europe. Instances of moderate to severe disease in association with this variant have also been reported (PMID: 17360088, 9482576, 11328279, 10354117). ClinVar contains an entry for this variant (Variation ID: 1513). Invitae Evidence Modeling of protein sequence and biophysical properties (such as structural, functional, and spatial information, amino acid conservation, physicochemical variation, residue mobility, and thermodynamic stability) indicates that this missense variant is not expected to disrupt PKLR protein function with a negative predictive value of 80%. Experimental studies have shown that this variant moderately affects PKLR function (PMID: 11960989). For these reasons, this variant has been classified as Pathogenic. (less)

Observation 1

Collection method: clinical testing
Allele origin: germline
Affected status: unknown

Pathogenic
(Feb 23, 2026)

C Contributing to aggregate classification

★☆☆☆
(Ambry Variant Classification Scheme 2023)

Inborn genetic diseases

Ambry Genetics
Accession: SCV007557274.1
First in ClinVar: Apr 25, 2026
Last updated: Apr 25, 2026

Publications:

PubMed: 11328279, 11960989, 18759866, 28133914, 29519373, 32036089, 9827908

Comment:

The c.1456C>T (p.R486W) alteration is located in exon 10 (coding exon 10) of the PKLR gene. This alteration results from a C to T substitution at nucleotide position 1456, causing the arginine (R) at amino acid position 486 to be replaced by a tryptophan (W). Based on data from gnomAD, the T allele has an overall frequency of 0.295% (833/282498) total alleles studied. The highest observed frequency was 0.812% (204/25114) of European (Finnish) alleles. This variant has been identified in the homozygous state and/or in conjunction with other PKLR variant(s) in individual(s) with features consistent with PKLR-related pyruvate kinase deficiency; in at least one instance, the variants were identified in trans (Zanella, 2001; Kedar, 2009; Jaouani, 2017; Svidnicki, 2018; Jamwal, 2020). This amino acid position is not well conserved in available vertebrate species. This alteration is predicted to be deleterious by in silico analysis. Based on the available evidence, this alteration is classified as pathogenic. (less)

Observation 1

Collection method: clinical testing
Allele origin: germline
Affected status: unknown

Pathogenic
(Feb 19, 2026)

C Contributing to aggregate classification

★☆☆☆
(DASA Assertion Criteria)

not provided

Dasa
Accession: SCV007598459.1
First in ClinVar: May 30, 2026
Last updated: May 30, 2026

Publications:

Classification (Last evaluated)	Review status (Assertion criteria)	Condition	Submitter	Collapse all rows

PubMed: 11054094, 11960989, 17574881, 27871768, 32974842, 9057665, 9657767, 9827908

Comment:

NM_000298.6(PKLR):c.1456C>T (p.Arg486Trp) is a missense variant that results in the substitution of arginine with tryptophan. This variant has been observed in affected individuals with related phenotype in a genotype context consistent with recessive disease (PMID: 11960989; PMID: 9057665; PMID: 9657767; PMID: 9827908; PMID: 11054094). Functional evidence supports a deleterious effect on the gene or gene product (PMID: 11960989; PMID: 9057665; PMID: 9657767; PMID: 9827908; PMID: 11054094). This variant has been recurrently observed in individuals with related phenotype (PMID: 11960989; PMID: 9057665; PMID: 9657767; PMID: 9827908; PMID: 11054094). Multiple computational predictions support a deleterious effect on the gene or gene product. The variant is present at low frequency in population datasets. Based on the available data, this variant is classified as pathogenic. (less)

Observation 1

Collection method: clinical testing
Allele origin: germline
Affected status: unknown

Pathogenic
(Oct 16, 2025)

C Contributing to aggregate classification

★☆☆☆
(Variantyx Assertion Criteria 2022)

Pyruvate kinase deficiency of red cells
(Autosomal recessive inheritance)

Variantyx, Inc.
Accession: SCV007600386.1
First in ClinVar: Jun 06, 2026
Last updated: Jun 06, 2026

Publications:

PubMed: 11328279, 1196098, 18759866, 28133914, 9827908

Comment:

This is a nonsynonymous variant in the PKLR gene (OMIM: 609712). Pathogenic variants in this gene have been associated with autosomal recessive pyruvate kinase deficiency. This variant has been reported in many unrelated affected individuals (PMID: 9827908, 18759866, 11328279, 28133914) (PS4_Very_Strong). Functional studies have shown that this variant alters PKLR protein function (PMID: 11960989) (PS3_Supporting), and multiple computational algorithms predict a deleterious effect for this variant (REVEL score: 0.914) (PP3_Moderate). This variant has a 0.4553% maximum allele frequency in non-founder control populations (https://gnomad.broadinstitute.org/). Based on the current evidence, this variant is classified as pathogenic for autosomal recessive pyruvate kinase deficiency. (less)

Observation 1

Collection method: clinical testing
Allele origin: germline
Affected status: no

Pathogenic
(Feb 12, 2024)

C Contributing to aggregate classification

★☆☆☆
(ACMG Guidelines, 2015)

Pyruvate kinase deficiency of red cells
(Autosomal dominant inheritance)

Department of Pediatrics, Duzce University
Accession: SCV007602635.1
First in ClinVar: Jun 14, 2026
Last updated: Jun 14, 2026

Publications:

PubMed: 32974842

Comment:

Well-characterised missense variant p.(Arg486Trp) (R486W), one of the most common pathogenic PKLR alleles in pyruvate kinase deficiency, described as a known pathogenic variant with published computational and functional support for reduced pyruvate kinase activity (Canu et al., Mol Biol Rep 2020; DOI 10.1007/s11033-020-05836-2) (PS3). Recurrently identified in affected individuals (PS4); rare in population databases (PM2_supporting); deleterious in silico predictions (PP3). Applied ACMG/AMP criteria: PS3, PS4, PM2_supporting, PP3. Classification: Pathogenic. (less)

Observation 1

Collection method: research
Allele origin: germline
Affected status: yes

Pathogenic
(Dec 02, 2025)

C Contributing to aggregate classification

★☆☆☆
(ACMG Guidelines, 2015)

Pyruvate kinase deficiency of red cells

3billion
Accession: SCV002521511.2
First in ClinVar: Jun 05, 2022
Last updated: Jun 20, 2026

Publications:

PubMed: 16704447, 8483951

Classification (Last evaluated)	Review status (Assertion criteria)	Condition	Submitter	Collapse all rows

Comment:

The variant is observed at an extremely low frequency in the gnomAD v4.1.0 dataset (total allele frequency: 0.295%). Predicted Consequence/Location: Missense variant In silico tool predictions suggest damaging effect of the variant on gene or gene product [REVEL: 0.91 (>=0.6, sensitivity 0.68 and specificity 0.92); 3Cnet: 0.98 (> 0.75, sensitivity 0.96 and precision 0.92)]. The same nucleotide change resulting in the same amino acid change has been previously reported as pathogenic/likely pathogenic with strong evidence (ClinVar ID: VCV000001513 /PMID: 8483951 /3billion dataset). The variant has been reported to be in trans with a pathogenic variant as either compound heterozygous or homozygous in at least one similarly affected unrelated individual (PMID: 16704447). A different missense change at the same codon (p.Arg486Leu) has been reported to be associated with PKLR-related disorder (PMID: 16704447). Therefore, this variant is classified as Pathogenic according to the recommendation of ACMG/AMP guideline. (less)

Observation 1

Collection method: clinical testing
Allele origin: germline
Affected status: yes
Platform type: exome sequencing

Pathogenic
(Dec 01, 2025)

C Contributing to aggregate classification

★☆☆☆
(CeGaT Center For Human Genetics Tuebingen Variant Classification Criteria Version 2)

not provided

CeGaT Center for Human Genetics Tuebingen
Accession: SCV001746867.33
First in ClinVar: Jul 10, 2021
Last updated: Jul 27, 2026

Comment:

PKLR: PM3:Very Strong, PM5, PP4:Moderate, PM2:Supporting, PS3:Supporting (less)

Observation 1

Collection method: clinical testing
Allele origin: germline
Affected status: yes
Number of individuals with the variant: 8

Pathogenic
(Nov 25, 2024)

C Contributing to aggregate classification

★☆☆☆
(ACMG Guidelines, 2015)

PKLR-related disorders

Blueprint Genetics
Accession: SCV007762245.1
First in ClinVar: Sep 22, 2026
Last updated: Sep 22, 2026

Observation 1

Collection method: clinical testing
Allele origin: germline
Affected status: yes
Number of individuals with the variant: 5
Test name: Exome, Hematology, Multisystem Disorders, Single gene tests

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Pathogenic
(Mar 20, 2026)

C Contributing to aggregate classification

★☆☆☆
(ACMG Guidelines, 2015)

PKLR-related condition

PreventionGenetics, part of Exact Sciences
Accession: SCV005344843.2
First in ClinVar: Oct 08, 2024
Last updated: Oct 03, 2026

Comment:

The PKLR c.1456C>T variant is predicted to result in the amino acid substitution p.Arg486Trp. This variant is one of the most common disease causing PKLR variants and has been reported in the homozygous and compound heterozygous states in many individuals with pyruvate kinase deficiency (OMIM #266200; Baronciani et al. 1993. PubMed ID: 8483951; Zarza et al. 1998. PubMed ID: 9827908; Valentini et al. 2002. PubMed ID: 11960989; Kedar et al. 2008. PubMed ID: 18759866). Functional studies indicate this variant disrupts protein function (Valentini et al. 2002. PubMed ID: 11960989). This variant is reported in 0.81% of alleles in individuals of European (Finnish) descent in gnomAD. This variant is interpreted as pathogenic. (less)

Observation 1

Collection method: clinical testing
Allele origin: germline
Affected status: unknown

Pathogenic
(Jan 06, 2020)

★★★★

Adenosine triphosphate, elevated, of erythrocytes

Reproductive Health Research and Development, BGI Genomics

Classification (Last evaluated)	Review status (Assertion criteria)	Condition	Submitter	Collapse all rows

N Not contributing to aggregate classification

Accession: SCV001142310.1
First in ClinVar: Jan 12, 2020
Last updated: Jan 12, 2020

Comment:

NM_000298.5:c.1456C>T in the PKLR gene has an allele frequency of 0.008 in European (Finnish) subpopulation in the gnomAD database. This variant has been reported in multiple individuals with Pyruvate kinase deficiency, in a compound heterozygous state: 1190A>T/1456C>T, 1042-1044del/1456C>T, 992A>G/1456C>T, 1436G>A/1456C>T (PMID: 18759866); 1456C>T/1675C>T, 1456C>T/1010G>A, 1456C>T/1223C>T, 1456C>T/1070T>C, 1456C>T/721G>T(PMID: 9827908). Functional studies suggest that the p.Arg486Trp variant may be a mild mutation with reduced enzyme activity (PMID: 11960989).Taken together, we interprete this variant as Pathogenic/Likely pathogenic variant. ACMG/AMP Criteria applied: PM3_VeryStrong, PS3, PP4. (less)

Observation 1

Collection method: curation
Allele origin: germline
Affected status: unknown

Pathogenic
(-)

★★★★

not provided

Department of Pathology and Laboratory Medicine, Sinai Health System
Additional submitter:
Franklin by Genoox
Study: The Canadian Open Genetics Repository (COGR)
Accession: SCV001550687.1
First in ClinVar: Apr 13, 2021
Last updated: Apr 13, 2021

N Not contributing to aggregate classification

Comment:

The PKLR p.R486W variant was identified in over 30 individuals with pyruvate kinase deficiency and haemolytic anemia as a compound heterozygous variant; phenotypes ranged from mild to severe (Russo_2018_PMID:29396846; Jaouani_2017_PMID:28133914; Montllor_2016_PMID:27871768; Kedar_2009_PMID:18759866; Zanella_2001_PMID:11328279; Manco_2000_PMID:11054094; Zarza_1998_PMID:9827908; Baronciani_1993_PMID:8483951; Marcello_2008_PMID:18708292). The variant was identified in dbSNP (ID: rs116100695) and ClinVar (classified as pathogenic by Fulgent Genetics, ARUP Laboratories and Reproductive Health Research and Development, BGI Genomics, as likely pathogenic by Laboratory of Molecular Medicine, EGL Genetics and Knight Diagnostic Laboratories, Oregon Health and Services University, and as likely benign by Invitae). The variant was identified in control databases in 833 of 282498 chromosomes (2 homozygous) at a frequency of 0.002949 (Genome Aggregation Database March 6, 2019, v2.1.1). The variant was observed in the following populations: European (Finnish) in 204 of 25114 chromosomes (freq: 0.008123), South Asian in 113 of 30616 chromosomes (freq: 0.003691), European (non-Finnish) in 375 of 128890 chromosomes (freq: 0.002909), Other in 20 of 7218 chromosomes (freq: 0.002771), Ashkenazi Jewish in 28 of 10360 chromosomes (freq: 0.002703), Latino in 80 of 35424 chromosomes (freq: 0.002258) and African in 13 of 24926 chromosomes (freq: 0.000522), but was not observed in the East Asian population. Although the p.R486 residue is not conserved in mammals and other organisms, computational analyses (PolyPhen-2, SIFT, AlignGVGD, BLOSUM, MutationTaster) suggest that the variant may impact the protein. The variant occurs outside of the splicing consensus sequence and 1 of 4 in silico or computational prediction software programs (SpliceSiteFinder, MaxEntScan, NNSPLICE, GeneSplicer) predict a greater than 10% difference in splicing; this is not very predictive of pathogenicity. Functional data suggests that this variant results in decreased catalytic efficiency (Valentini_2002_PMID:11960989). In summary, based on the above information this variant meets our laboratoryâ€šÃ„Ã´s criteria to be classified as pathogenic. (less)

Observation 1

Collection method: clinical testing
Allele origin: unknown
Affected status: yes

Pathogenic
(Feb 01, 2009)

★★★★

ANEMIA, CONGENITAL, NONSPHEROCYTIC HEMOLYTIC, 2

OMIM
Accession: SCV000021733.3
First in ClinVar: Apr 04, 2013
Last updated: Oct 20, 2024

N Not contributing to aggregate classification

Publications:

PubMed: 11328279, 18759866, 9827908

Observation 1

Collection method: literature only
Allele origin: germline
Affected status: not provided
Comment on evidence:
Zarza et al. (1998) found that the most frequent mutation of the PKLR gene causing pyruvate kinase deficiency and hereditary nonspherocytic hemolytic anemia (CNSHA2; 266200) in Spain is a 1456C-T transition, resulting in an arg486-to-trp (R486W) substitution. The mutation was identified in approximately one-third of mutant alleles (7 of 22). Zanella et al. (2001) found that the R486W mutation appeared to be the most frequent mutation causing hemolytic anemia associated with PK deficiency in Italy. Kedar et al. (2009) identified the R486W mutation in 6 (16.7%) of 36 alleles in 18 Indian patients with PK deficiency. (less)

Citations for germline classification of this variant

Title	Author	Journal	Year	Link
Targeted Next Generation Sequencing and Diagnosis of Congenital Hemolytic Anemias: A Three Years Experience Monocentric Study.	Fermo E *et al.*	Frontiers in physiology	2021	PMID: 34093240
Six novel variants in the PKLR gene associated with pyruvate kinase deficiency in Argentinian patients.	Milanesio B *et al.*	Clinical biochemistry	2021	PMID: 33631127
Identification and in silico characterization of a novel PKLR genotype in a Turkish newborn.	Canu G *et al.*	Molecular biology reports	2020	PMID: 32974842
Next-Generation Sequencing-Based Diagnosis of Unexplained Inherited Hemolytic Anemias Reveals Wide Genetic and Phenotypic Heterogeneity.	Jamwal M *et al.*	The Journal of molecular diagnostics : JMD	2020	PMID: 32036089
Precision medicine integrating whole-genome sequencing, comprehensive metabolomics, and advanced imaging.	Hou YC *et al.*	Proceedings of the National Academy of Sciences of the United States of America	2020	PMID: 31980526
Interpretation of Genomic Sequencing Results in Healthy and Ill Newborns: Results from the BabySeq Project.	Ceyhan-Birsoy O *et al.*	American journal of human genetics	2019	PMID: 30609409
Precision medicine screening using whole-genome sequencing and advanced imaging to identify disease risk in adults.	Perkins BA *et al.*	Proceedings of the National Academy of Sciences of the United States of America	2018	PMID: 29555771
Novel mutations associated with pyruvate kinase deficiency in Brazil.	Svidnicki MCCM *et al.*	Hematology, transfusion and cell therapy	2018	PMID: 29519373
Multi-gene panel testing improves diagnosis and management of patients with hereditary anemias.	Russo R *et al.*	American journal of hematology	2018	PMID: 29396846
Molecular basis of pyruvate kinase deficiency among Tunisians: description of new mutations affecting coding and noncoding regions in the PKLR gene.	Jaouani M *et al.*	International journal of laboratory hematology	2017	PMID: 28133914

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Text-mined citations for rs116100695 LitVar 2

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